Integer points on a curve family

Problèmes ouverts — A. Dujella · 2026

Problem 4.9 (Dujella, 2026). Let $k \geqslant 3$ be an integer. Is it true that all integer points on the elliptic curve $y^2 = ((k-1)x+1)((k+1)x+1)(4kx+1)$ are given by $(x,y) \in \{(0,\pm 1), (16k^3 - 4k, \pm(128k^6 - 112k^4 + 20k^2 - 1))\}$?

Remark. The positive answer is known if the rank of the curve is equal to 1 (by Dujella (Acta Arith. **94** (2000), 87–101)) or $k \leqslant 5000$ (by Najman (Rocky Mountain J. Math. **40** (2010), 1979–2002)). Also if $k = 3n^2 + 2n - 2$ or $k = \frac{1}{2}(3m^2 + 5m)$ and the rank is equal to 2 (by Dujella (Acta Arith. **94** (2000), 87–101)).